Multiple Choice and short answer (5 points each)
In the following questions, pick the <u>most appropriate</u> selection for the answer.

1. Which of the following models the permitted events of the objects in a class:

A. Use case diagrams
B. Event diagrams
C. Sequence diagrams
D. State diagrams
E. Activity diagrams

The events in the use case diagram, event diagram, and sequence diagram permits the class object. The events of the object are highlighted in the diagram.

2. Look at the following code.

```
Line  1    public class ClassA
Line  2    {
Line  3        public ClassA() {}
Line  4        public int method1(int a){}
Line  5        public int method2(int b){}
Line  6    }

Line  7    public class ClassB extends ClassA
Line  8    {
Line  9        public ClassB(){}
Line 10        public int method2(double c){}
Line 11    }

Line  12   public class classC extends ClassA
Line  13   {
Line  14       public ClassC(){}
Line 15        public int method1(int b){}
Line 16        public int method2(double c){}
Line 17    }
```

Assume the classes ClassA , ClassB , ClassC are defined similar to the above. If the reference variables var1, var2, var3, and var4 are declared as:

```
ClassA var1 = new ClassA();      ClassB var2  = new ClassB();
ClassC var3 = new ClassC();      ClassB var4 = new ClassB();
```

Which of the following lines is indicative of polymorphism in Java?

A. var3 = var1;
B. var2 = var3;
C. var4 = var3;
D. var4 = var1;
E. var1 = var3;

parent = child

3. The life lines are part of which one of the following in a UML model?

A. Domain class models
B. Use case diagrams
C. State diagrams
D. Sequence diagrams
E. Activity Diagram

4. All fields declared in an interface:

A. Have protected access
B. Must be initialized in the class implementing the interface
C. Have public access
D. Are final and static
E. Interfaces have no fields, only method headers

5. Pick the best choice for being true (A, B, C, D, or E). An action listener for a JButton component must:

i. be an inner class.
ii. implement the ActionListener interface.
iii. have a method named actionPerformed.

A. (i), (ii), and (iii).
B. Both (i) and (ii)
C. (i) necessary, but not (ii)
D. Both (ii) and (iii)
E. Only (iii)

6. Look at the following code, and assume it compiles/runs.

```
public interface Animal {
   public void reproduce();
   public void travel();
}

public class Ml implements Animal {
   protected double spd;
   public void reproduce() { system.out.println("I give birth."); }
   public void travel() { system.out.println("I run fast"); }
}

public class Rl implements Animal {
   protected double spd;
   public void reproduce() { system.out.println("I lay eggs."); }
   public void travel() { system.out.println("I crawl"); }
}

public class Dg extends Ml {
   public void eat() { system.out.println("I eat meat."); }
   public void reproduce() { system.out.println("No way!"); }
}
```

Which of the following choices will *not* be correct?

A. `Animal a = new Rl();  a.reproduce();`
B. `Ml ml = new Dg();  ml.reproduce();`
C. `Animal a = new Animal();  a.reproduce();`
D. `Dg d = new Dg();  d.travel();`
E. `Dg d = new Dg();  d.spd = 10.0;`

7. An `Insect` class is to be written containing the following data fields:
`age`, which will be initialized to 0 when an insect is constructed,
`nextAvailableID`, which will be initialized to 0 outside the constructor and incremented each time an `insect` is constructed,
`idNum`, which will be initialized to the current value of the `nextAvailableID` when an `insect` is constructed,
`position`, which will be initialized to the position in a garden the `insect` is placed,
`direction`, which will be initialized to the direction the `insect` is facing when it is placed in the garden,

Which variable in the `Insect` class should be static?

A. `Age`
B. `position`
C. `direction`
D. `nextAvailableID`
E. `idNum`

8. (10 points) Fill in the blanks so that the following draws a Frame containing "Hello":

```
import javax.swing.*;
import java.awt.*;

public class helloFrame ___extends JFrame___ {


      public helloFrame() {

            __super____("Hello");

            setSize(200,150);
            setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
            __setVisible__(true);
      }

      public void ____paint____ (_Graphics g_ )
      {
            Super.paintComponet(g);

            g.drawString("Hello", 50, 50 );
      }
}

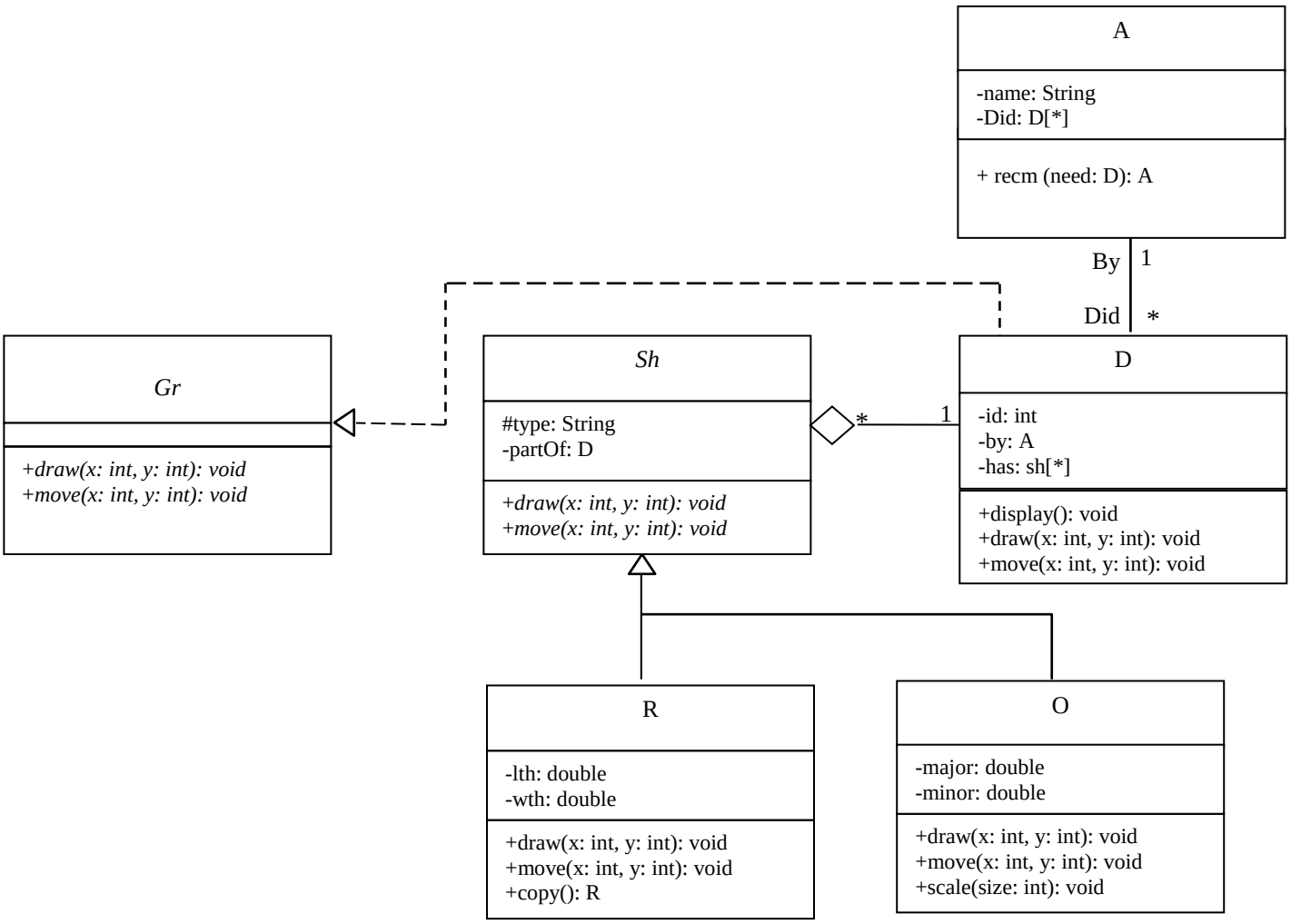
public class Tester  {
      public static void main ( String[] args )   {
            helloFrame frm = new helloFrame();
      }
}
```

9. A program is to be written that simulates bumper cars in a video game. The cars move on a grid and are located on grid points `(x,y)`, where `x` and `y` are integers between -20 and 20. A bumper car moves in a random direction, either left, right, up, or down. If it reaches a grid boundary it reverses direction. It also reverses direction if it is about to collide with another bumper car. One step (st) of the simulation moves each car in the grid one step. After a bumper car has reversed direction twice, its turn is over and the next car gets to move. Your program shall be able to add bumper cars and run the simulation.

The programmer decided to design a `BumperCar` class which should have an identification number, position on the grid, and current direction. Which of the following methods is not suitable to be implemented in the class `BumperCar` ?

A. `public Boolean atBoundary()` // Returns true if BumperCar is at boundary
B. `public int selectRandomDirection()` // randomly select a direction
C. `public void reverseDirection()` // change the car direction to opposite of current direction;
D. `public void move()` // Take turn to move, and stop moving after two changes in direction;
E. `public void update()` // update the grid to reflect new position after each simulation step;

2. (25 points) You are given the following design in UML. Show the java declarations for implementing this design. Assume that the concrete methods have either empty definitions (i.e. their definition is {}), or only create and return an object of the appropriate type (i.e. calling the appropriate default constructor).



```
interface Gr {
  void draw(int x,int y);
  void move(int x,int y);
}

class A {
  private String name;
  private D[] Did;

  public A recm(D need){
    return  new A();
  }
}

class D implements Gr{
  private int id;
  private A by;
  private Sh[] has;
  public void display(){}

  @Override
    public void draw(int x, int y) { }

  @Override
    public void move(int x, int y) { }
}

class Sh extends A {
  protected String type;
  private D partOf;

  public void draw(int x, int y) { }
  public void move(int x, int y) { }
}

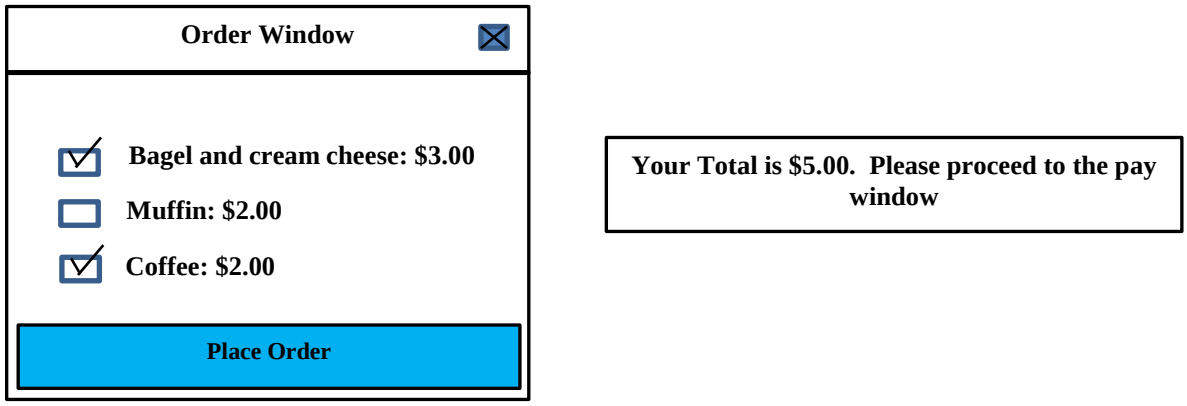
class R extends Sh {
  private double ith;
  private double wth;

  public void draw(int x, int y) { }
  public void move(int x, int y) { }
  public  R copy() {
    return  new R();
  }
}

class O extends Sh {
  private double major;
  private double minor;

  public void draw(int x, int y) { }
  public void move(int x, int y) { }
  public void scale(int size){ }
}
```

3. (25 points)
You have a user interface concept for a fast food application (see below). The user shall be allowed to select from the three choices, in whatever combination he/she wants, for example Bagel and cheese with coffee, or just coffee. Once the user has made his/her selections, he places his order by pressing the Place Order button. A pop-up as shown on the right must come up after the Place Order is pressed to indicate the total amount and ask user to proceed to the pay window. Your task is to create the user interface as shown.



Complete the code below to show how you will create the order window with boxes that a user can check to describe what he/she wants to order. Places that are indicated as "**.....**" may need multiple lines to complete.

Hint: The following line can be used to produce a pop-up with the desired content:

```
JOptionPane.showMessageDialog(null, "Your Total is $" + totalOrder + ". Please proceed to the pay window.");
```

```
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;

public class OrderWindow extends JFrame {

       private JPanel orderPanel;
       private JCheckBox bagel = new JCheckBox("Bagel and Cream Cheese");
       private JCheckBox muffin = new JCheckBox("muffin");
       private JCheckBox coffee = new JCheckBox("coffee");
       private JButton order = new JButton("Place Order");

       public OrderWindow() {
             super("Order Window)";

             setSize(300,300);
             setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
             buildPanel();    // build order panel

             add(_orderPanel___);

               setVisible(true)      ;

          }
```

```
        private void buildPanel()
           {
                orderPanel = new JPanel()
                ____________________________;

                orderPanel.setLayout(new GridLayout(4, 1));
                ......        //finish the buildPanel method
                                        orderPanel.add(bagel);
           }                            orderPanel.add(muffin);
                                        orderPanel.add(coffee);
    //finish the rest of OrderWindow    orderPanel.add(order);
        ......
        ......


}

public class PlaceOrder {

        public static void main(String[] args) {
                new OrderWindow();
        }
}                                       order.addActionListener(new ActionListener() {

                                                @Override
                                             public void actionPerformed(ActionEvent e) {
                                                    double total=0;
                                                  if(bagel.isSelected())
                                                        total+=3;
                                                    if(muffin.isSelected())
                                                        total+=2;

                                                    if(coffee.isSelected())
                                                        total+=2;

                                        JOptionPane.showMessageDialog(null, "Your total is :
                                        $"+total+"\n Please proceed to the pay window")
                                        }
                                        });
```